Main results - MASA03

Probability spaces

Definition. A *probability space* is a triple $(\Omega, \mathcal{F}, P)$ where:

- Ω: *sample space* (set of outcomes).
- $\mathcal{F} \subseteq 2^{\Omega}$: *$\sigma$-algebra* of events: $\Omega \in \mathcal{F}$; if $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$; if $A_1, A_2, \dots \in \mathcal{F}$ then $\bigcup_{n\geq 1} A_n \in \mathcal{F}$.
- $P : \mathcal{F} \to [0,1]$: *probability measure* satisfying the axioms:
  (i) Non-negativity: $P(A) \geq 0$ for all $A \in \mathcal{F}$.
  (ii) Normalization: $P(\Omega) = 1$.
  (iii) σ-additivity: if $A_i \in \mathcal{F}$ are pairwise disjoint, $P\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \sum_{i=1}^{\infty} P(A_i)$.

Immediate consequences (useful): $P(\varnothing) = 0$, $\quad P(A^c) = 1 - P(A)$, $\quad A \subseteq B \Rightarrow P(A) \leq P(B)$, $\quad$ if $A \cap B = \varnothing$ then $P(A \cup B) = P(A) + P(B)$.

0.1 Random Variable

Definition. A function $X : \Omega \to \mathbb{R}$ is a *random variable* on the probability space $(\Omega, \mathcal{F}, P)$ if it satisfies the condition of $\mathcal{F}$-measurability:

$$\text{For every Borel set } B \in \mathcal{B}(\mathbb{R}), \quad X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}$$

Required Conditions

The crucial condition is that the pre-image of any set B for which a probability is defined in $\mathbb{R}$ must be an event in the σ-algebra $\mathcal{F}$. The probability measure P on $\mathcal{F}$ itself is not involved in defining X as a random variable, only the structure of $\mathcal{F}$.

Law of Total Probability (LTP)

If $A_1, \dots, A_k$ form a partition of Ω (pairwise disjoint, $\bigcup_i A_i = \Omega$), then for any event B,

$$P(B) = \sum_{i=1}^{k} P(B \mid A_i)\, P(A_i).$$

(Countable version holds similarly.)

Bayes' Rule

For events A_j and B with $P(B) > 0$,

$$P(A_j \mid B) = \frac{P(B \mid A_j)\,P(A_j)}{\sum_i P(B \mid A_i)\,P(A_i)}.$$

The denominator is $P(B)$ from the LTP over the partition $\{A_i\}$.

1 Joint distributions and tools

1.1 Joint density and normalization

A pair (X, Y) has joint density $f_{X,Y}(x, y) \geq 0$ with

$$\iint_{\mathbb{R}^2} f_{X,Y}(x, y)\,dx\,dy = 1.$$

1.2 Marginals

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\,dy, \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\,dx.$$

1.3 Independence

X and Y are independent iff

$$f_{X,Y}(x, y) = f_X(x)\,f_Y(y) \quad \text{(almost everywhere on the support)}.$$

1.4 Conditional density

For $f_X(x) > 0$,

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

1.5 Cumulative distribution function (cdf)

For any random variable Z,

$$F_Z(z) = \Pr(Z \leq z), \qquad \Pr(a < Z \leq b) = F_Z(b) - F_Z(a).$$

If Z has a density f_Z, then

$$f_Z(z) = \frac{d}{dz} F_Z(z).$$

1.6 Product variable via the cdf method

For $Z = XY$ with (X, Y) supported on $[a, b] \times [c, d] \subseteq [0, \infty)^2$,

$$F_Z(z) = \Pr(XY \leq z) = \iint_{\{(x,y) \in [a,b] \times [c,d] :\, xy \leq z\}} f_{X,Y}(x, y)\,dx\,dy,$$

and $f_Z(z) = \dfrac{d}{dz} F_Z(z)$ on the interior of the support of Z.

2 Expectation and related results

2.1 Definition

For a random variable X with pdf f_X or pmf p_X,

$$E[X] = \begin{cases} \displaystyle\sum_x x\, p_X(x), & \text{(discrete)}, \\ \displaystyle\int_{-\infty}^{\infty} x\, f_X(x)\, dx, & \text{(continuous)}. \end{cases}$$

More generally (Law of the Unconscious Statistician, LOTUS),

$$E[g(X)] = \begin{cases} \displaystyle\sum_x g(x)\, p_X(x), \\ \displaystyle\int g(x)\, f_X(x)\, dx. \end{cases}$$

2.2 Conditional expectation and total laws

For densities,

$$E[X \mid Y = y] = \int x\, f_{X|Y}(x \mid y)\, dx, \qquad f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

Law of total expectation (tower property):

$$E[X] = E\big(E[X \mid Y]\big).$$

Law of total probability (continuous form): For any event A,

$$P(A) = \int P(A \mid Y = y)\, f_Y(y)\, dy \quad \text{or, for discrete } Y, \quad P(A) = \sum_y P(A \mid Y = y)\, P(Y = y).$$

This expresses the overall probability $P(A)$ as the average (expectation) of the conditional probabilities $P(A \mid Y = y)$ weighted by how likely each y is. Equivalently,

$$P(A) = E\big[P(A \mid Y)\big].$$

Hence, probabilities are special cases of expectations:

$$P(X < Y) = E\big[P(X < Y \mid X)\big].$$

2.3 Variance and Covariance

$$\mathrm{Var}(X) = E[X^2] - \big(E[X]\big)^2$$
$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

2.3.1 Law of Total Variance (Conditional Version)

$$\mathrm{Var}(X) = E[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(E[X \mid Y])$$

2.3.2 Relationship Between Cov and Var

$$\mathrm{Var}(X) = \mathrm{Cov}(X, X)$$

2.3.3 Correlation Coefficient (ρ)

The correlation coefficient is the standardized covariance.

$$\rho_{X,Y} = \mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)}\sqrt{\mathrm{Var}(Y)}} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Identity: The identity for correlation is its range:

$$-1 \leq \rho_{X,Y} \leq 1$$

Covariance in terms of Correlation:

$$\mathrm{Cov}(X, Y) = \rho_{X,Y} \sigma_X \sigma_Y$$

3 Common Probability Distributions

Below are several standard distributions frequently used in probability and statistics, with their probability mass or density functions and key characteristics.

3.1 Discrete distributions

Binomial distribution

$$X \sim \mathrm{Bin}(n, p)$$

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

$$E[X] = np, \qquad \mathrm{Var}(X) = np(1-p).$$

Bernoulli distribution

$$X \sim \mathrm{Bern}(p)$$

$$P(X = 1) = p, \quad P(X = 0) = 1 - p.$$

$$E[X] = p, \qquad \mathrm{Var}(X) = p(1-p).$$

Poisson distribution

$$X \sim \mathrm{Poisson}(\lambda)$$

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

$$E[X] = \lambda, \qquad \mathrm{Var}(X) = \lambda.$$

3.2 Continuous distributions

Uniform distribution

$$X \sim \mathrm{Unif}(a, b)$$

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

$$E[X] = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}.$$

Exponential distribution

$$X \sim \mathrm{Exp}(\lambda)$$

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \ge 0, \\ 0, & \text{otherwise.} \end{cases}$$

$$E[X] = \frac{1}{\lambda}, \qquad \mathrm{Var}(X) = \frac{1}{\lambda^2}.$$

Memoryless property: $P(X > s + t \mid X > s) = P(X > t)$.

Normal (Gaussian) distribution

$$X \sim N(\mu, \sigma^2)$$

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R}.$$

$$E[X] = \mu, \qquad \mathrm{Var}(X) = \sigma^2.$$

Standard Normal distribution

$$Z \sim N(0, 1)$$

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \qquad F_Z(z) = \Phi(z) = \int_{-\infty}^{z} f_Z(t)\, dt.$$

Gamma distribution

$$X \sim \mathrm{Gamma}(\alpha, \lambda)$$

$$f_X(x) = \begin{cases} \dfrac{\lambda^\alpha}{\Gamma(\alpha)}\, x^{\alpha-1} e^{-\lambda x}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

$$E[X] = \frac{\alpha}{\lambda}, \qquad \mathrm{Var}(X) = \frac{\alpha}{\lambda^2}.$$

Special case: $\mathrm{Exp}(\lambda) = \mathrm{Gamma}(1, \lambda)$.

3.3 Summary Table

Distribution	Support	Mean	Variance
Bernoulli(p)	$\{0, 1\}$	p	$p(1-p)$
Binomial(n, p)	$\{0, \ldots, n\}$	np	$np(1-p)$
Poisson(λ)	$\mathbb{N}_0$	λ	λ
Uniform(a, b)	$[a, b]$	$(a+b)/2$	$(b-a)^2/12$
Exponential(λ)	$[0, \infty)$	$1/\lambda$	$1/\lambda^2$
Normal(μ, σ^2)	$\mathbb{R}$	μ	σ^2
Gamma(α, λ)	$[0, \infty)$	α/λ	α/λ^2

4 Limit Theorems: LLN and CLT

4.1 Law of Large Numbers (LLN)

Let $X_1, X_2, \ldots$ be i.i.d. random variables with $E[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Define the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Weak Law of Large Numbers (WLLN)

$$\bar{X}_n \xrightarrow{P} \mu \quad \text{as } n \to \infty.$$

That is, for every $\varepsilon > 0$,

$$P(|\bar{X}_n - \mu| > \varepsilon) \longrightarrow 0.$$

Interpretation: The sample mean converges in probability to the true mean μ.

Strong Law of Large Numbers (SLLN)

$$\bar{X}_n \xrightarrow{\text{a.s.}} \mu \quad \text{as } n \to \infty.$$

That is,

$$P\Big(\lim_{n\to\infty} \bar{X}_n = \mu\Big) = 1.$$

Interpretation: The sample mean converges almost surely (with probability 1) to the true mean μ.

Difference between weak and strong forms: Both state that $\bar{X}_n$ approaches μ as n grows, but:

- **Weak LLN:** convergence in probability (approximate closeness for large n).
- **Strong LLN:** almost sure convergence (the entire sequence converges for almost every outcome).

4.2 Central Limit Theorem (CLT)

Theorem (Lindeberg-Lévy CLT). Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with

$$E[X_i] = \mu, \qquad \mathrm{Var}(X_i) = \sigma^2 < \infty.$$

Then as $n \to \infty$,

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xRightarrow{d} N(0,1),$$

where $\xRightarrow{d}$ denotes convergence in distribution.

Interpretation: For large n, the sample mean $\bar{X}_n$ is approximately normal:

$$\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

Hence, the CLT justifies using the normal distribution to approximate sums or averages of many independent random variables.

Practical consequence: For large n,

$$P\left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq z\right) \approx \Phi(z),$$

where $\Phi(z)$ is the standard normal cdf. This result underpins confidence intervals and hypothesis testing.